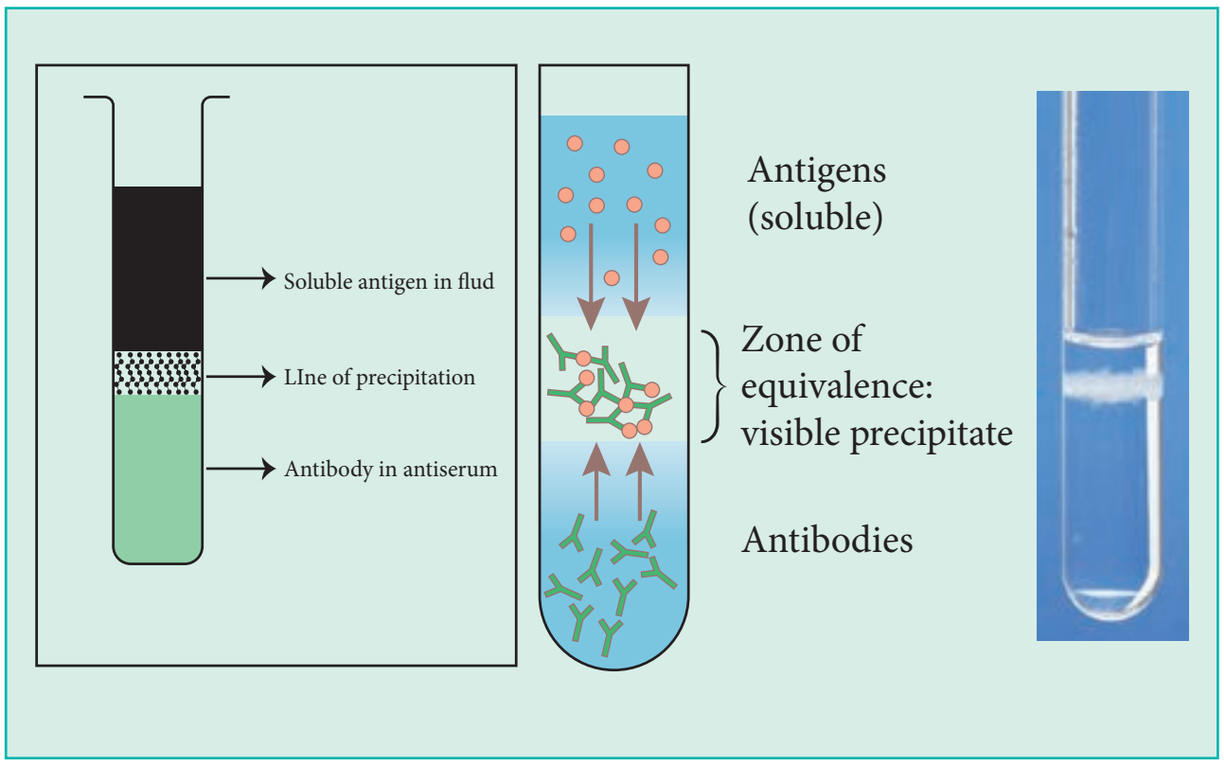
10.6.1. Precipitation

The interaction between an antibody and a soluble antigen in aqueous solution forms a lattice that eventually develops a visible precipitation. Antibody that forms precipitation is known as precipitin. This process is called as precipitation reaction. Formation of an Ag-Ab lattice depends on the valency of both antigen and antibody. Zone of equivalence is a point at which the maximum precipitation occurs. This reaction is widely used in several immunological techniques.

Ring test: In this test, over a column of antiserum (antibody), the antigen solution is layered. A precipitate forms at the junction of the two liquids.



10.6.2. Agglutination

The interaction between antibody and a particulate antigen results in visible clumping called agglutination. Antibodies that produce such reactions are called agglutinins. Agglutination reactions are similar in principle to precipitation reactions; they depend on the cross linking of polyvalent antigens. Just as an excess of antibody inhibits precipitation reactions, such excess can also inhibit agglutination reactions; this inhibition is called the prozone effect.

Haemagglutination (Slide Agglutination)

In typing for the ABO antigens, RBCs are mixed on a slide with antisera to the A or B blood-group antigens. If the antigen is present on the cells, they agglutinate, forming a visible clump on the slide. Determination of which antigens are present on donor and recipient RBCs is the basis for matching blood types for transfusions. At neutral pH, red blood cells are surrounded by a negative ion cloud that makes the cells repel one

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